

Research Summary

Safwat Hassan
 Assistant Professor
 University of Toronto, ON, Canada
 safwat.hassan@utoronto.ca

Mobile applications (*apps*) are becoming an integral part of our daily activities. The mobile market is continuously growing, with an expected \$156 billion in revenue by 2023 [4]. With the mobile market's highly competitive nature, app developers need to avoid deploying buggy releases and continuously deploy high-quality releases that satisfy user needs. Over the years, the growing mobile apps market with millions of apps and billions of yearly downloads generates comprehensive data. Mining mobile app market data (e.g., analyzing user reviews) can help app developers improve their apps' perceived quality.

My research focuses on applying data mining techniques to the emerging software repositories (e.g., the app market data) to improve the perceived quality of software products. In the last six years, I have been working on providing techniques that can help app developers and store owners improve mobile apps' perceived quality in the following directions.

1. Analyzing Users' Feedback

We proposed approaches that help app developers proactively spot buggy releases and deploy fixes for such buggy releases [20, 32, 25]. Moreover, we proposed approaches to spot user complaints that may lead to user churn (i.e., migrating to competitor products) [14] and prioritize the required features based on the app's competitors [6, 21, 7]. We proposed an approach to identify the topics of the raised issues in the bug reports [5]. We also proposed approaches to help app developers design mobile apps with a high-quality user interface (UI) [11].

2. Analyzing Developers' Common Practices

We studied the common practices of delivering main features in mobile apps, such as the patterns of writing release notes [33], integrating third-party libraries [1, 2, 31, 30, 18], and managing deep learning frameworks [8]. We analyzed the practices of integrating Large Language models (LLMs) in mobile apps [24] and how users perceive the adoption of LLMs in mobile apps [3]. Our work shows the challenges and promises of adopting LLM in mobile apps [3, 24]. We also identified the common mistakes that lead to emergency fixes in mobile apps [22]. This work highlighted the importance of analyzing the store data on a large scale to automatically identify the deployed issues and recommend solutions to the identified issues.

3. Studying the Interactions Between Users and Developers

Responding to user reviews can improve the ratings of an app. However, with many daily user reviews, it is not practical to respond to all reviews. Hence, we proposed an automatic approach to predict reviews that are most likely to get app developers' responses [23]. We also studied developers' interactions using development communication platforms, such as the Gitter platform [13]. We proposed an approach that identifies discussion threads in communication platforms. Project maintainers can benefit from our work to spot the commonly repeated discussions and use such discussions to improve

the documentation of their projects. To ease the replication of our study, we shared our dataset in our replication package [13].

4. Improving the Performance of Software Systems

We collaborate with IBM to provide approaches to enhance the performance of software systems [35, 15, 34]. We also proposed different approaches to improve the design and maintenance of software systems [26, 27, 28].

5. Adopting Mobile Apps for Healthy Life

We collaborated with medical practitioners to build mobile apps that adopt machine learning techniques to predict medical issues in users (patients). In addition, we proposed approaches that provide proactive recommendations for patients and practitioners to avoid severe consequences of the predicted medical issues [12, 29].

6. Towards Rapid Releases of Software Systems

We analyze the build results of 924,616 builds of 588 GitHub projects. We identified best practices to improve the build performance of open-source software systems [17]. We also study the practices of adopting Continuous Integration and Continuous Delivery (**CI/CD**) in mobile apps [16, 9, 36]. In addition, we proposed “LogSieve” as an approach to reduce the generated CI logs in mobile apps [10]. Our work aims to identify the promises and challenges of adopting CI/CD practices in mobile apps [19, 36]. Our work shows variations in the impact of adopting CI practices on the success of mobile apps [36].

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